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English

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN) path to success

# CLASSROOM CONTACT PROGRAMME

(Academic Session : 2024 - 2025)

Test Pattern

NEET(UG)
MAJOR
17-03-2025

**PRE-MEDICAL : LEADER & ACHIEVER COURSE PHASE - MLB, MLK, MLM, MLP, MLQ, MLR, MAZE, MAZF, MAZG, MAZH, MAZM, MAZP, MAZQ, MAZR, MAZU, MAAZ, MAAW, SPARK PRO**

**Test Booklet Code**

# L2

This Booklet contains **24** pages.

**Do not open this Test Booklet until you are asked to do so.**

***Important Instructions :***

1. The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on ORIGINAL Copy carefully with **blue/black** ball point pen only.
2. The test is of **3 hours** duration and this Test Booklet contains **180** questions. Each question carries **4** marks. For each correct response, the candidate will get **4** marks. For each incorrect response, **one mark** will be deducted from the total scores. The maximum marks are **720**.
3. Use **Blue/Black Ball Point Pen only** for writing particulars on this page/marking responses on Answer Sheet.
4. Rough work is to be done in the space provided for this purpose in the Test Booklet only.
5. On completion of the test, the candidate **must hand over the Answer Sheet (ORIGINAL and OFFICE Copy) to the Invigilator** before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
6. The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Form No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
7. Use of white fluid for correction is **NOT** permissible on the Answer Sheet.
8. Each candidate must show on-demand his/her Allen ID Card to the Invigilator.
9. No candidate, without special permission of the Invigilator, would leave his/her seat.
10. The candidates should not leave the Examination Hall without handing over their Answer Sheet to the Invigilator on duty and sign (with time) the Attendance Sheet **twice. Cases, where a candidate has not signed the Attendance Sheet second time, will be deemed not to have handed over the Answer Sheet and dealt with as an Unfair Means case**.
11. Use of Electronic/Manual Calculator is prohibited.
12. The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Room/Hall. All cases of unfair means will be dealt with as per the Rules and Regulations of this examination.
13. **No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.**
14. The candidates will write the Correct Test Booklet Code as given in the Test Booklet/Answer Sheet in the Attendance Sheet.

Name of the Candidate (in Capitals) : ______________________

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**YOUR TARGET IS TO SECURE GOOD RANK IN PRE-MEDICAL 2025**

## SUBJECT : CHEMISTRY

**Topic :** SYLLABUS-2.

**1.** In chromium atom number of electrons which have n = 2 and $\ell$ = 1 is :-

(1) 6 (2) 4 (3) 2 (4) 1

**2.** Among the orbitals $3d_{xy}$, $2p_x$, 3s and 2s, which one has maximum number of nodal planes and radial nodes respectively :-

(1) 3s and 2s

(2) $3d_{xy}$ and 2s

(3) $2p_x$ and 3s

(4) $3d_{xy}$ and 3s

**3.** In the energy level sequence for a multi-electron atom, 4f-orbital lies between :-

(1) 6s and 5d

(2) 5p and 5d

(3) 4p and 5p

(4) 5d and 6d

**4.** The first emission line of Balmer series for H spectrum has the wave no. equal to ?

(1) $\frac{9R}{40}$ (2) $\frac{7R}{144}$

(3) $\frac{36}{5R}$ (4) $\frac{5R}{36}$

**5.** On the basis of Bohr's model, the radius of 3rd orbit is :-

(1) Equal to the radius of first orbit

(2) Three times the radius of first orbit

(3) Nine times the radius of first orbit

(4) Five times the radius of first orbit

**6.** The ratio of the radii of first orbits of H, $He^+$ and $Li^{2+}$ is :-

(1) 1 : 2 : 3 (2) 6 : 3 : 2

(3) 1 : 4 : 9 (4) 9 : 4 : 1

**7.** Match the following :

| | Column-I | | Column-II |
|---|---|---|---|
| i | Aufbau principle | a | $\lambda = h/(mv)$ |
| ii | de broglie | b | $\lambda = hc/\Delta E$ |
| iii | Hund's rule | c | Max. unpaired $e^-$ |
| iv | Planck's law | d | Electronic configuration |

(1) (i) d (ii) b (iii) c (iv) a

(2) (i) c (ii) a (iii) d (iv) b

(3) (i) d (ii) a (iii) c (iv) b

(4) (i) c (ii) b (iii) d (v) a

**8.** The e/m ratio is maximum for :-

(1) $D^+$ (2) $He^+$ (3) $H^+$ (4) $He^{2+}$

**9.** The de-broglie wavelength associated with particle of mass $10^{-6}$ kg with a velocity of 10 $ms^{-1}$ is :-

(1) $6.63 \times 10^{-22}$ m (2) $6.63 \times 10^{-29}$ m

(3) $6.63 \times 10^{-31}$ m (4) $6.63 \times 10^{-34}$ m

**10.** Distance between $2^{nd}$ and $3^{rd}$ shell of H-atom is :-

(1) 1.2 Å (2) 2.65 Å

(3) 4.30 Å (4) 1.65 Å

**11.** Which of the following set of quantum number is not possible :-

(1) n = 3, $\ell$ = 2, m = +1, s = −1/2

(2) n = 4, $\ell$ = 1, m = +2, s = +1/2

(3) n = 4, $\ell$ = 3, m = 0, s = +1/2

(4) n = 3, $\ell$ = 1, m = 0, s = −1/2

**12.** The total number of atomic orbitals in the third energy level of an atom is

(1) 9 (2) 8 (3) 18 (4) 32

**13.** Select the **incorrect** statement about $8^{th}$ period ?

(1) The maximum no of electrons in this period sub shell = 50

(2) The maximum no. of elements in this period = 50

(3) The types of subshell in this period = 5

(4) The maximum no of orbitals in new sub shell in this period = 10

**14.** The compound of vanadium with chlorine has magnetic moment 1.73 B. M. The vanadium chloride has the formula :-

(1) $VCl_2$ (2) $VCl_3$

(3) $VCl_4$ (4) $VCl_5$

**15.** $[Rn]6d^2, 7s^2$ is the configuration of :-

(1) Ac, d-block (2) Th, f-block

(3) Pa, f-block (4) Hf, d-block

**16.** On moving in a transition series from left to right increase in the $Z_{eff}$ is very less due to

(1) Some Ionic potential

(2) Abnormal behaviour of d-block series

(3) poor shielding effect of d-orbitals

(4) Shielding effect of (n − 1) shell is more.

**17.** Match **List-I** with **List-II** :

| | List-I Elements | | List-II Atomic radii (pm) |
|---|---|---|---|
| (a) | O | (i) | 88 |
| (b) | C | (ii) | 74 |
| (c) | B | (iii) | 66 |
| (d) | N | (iv) | 77 |

Choose the **correct answer** from the options given below :

(1) (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)

(2) (a)-(i), (b)-(iv), (c)-(iii), (d)-(ii)

(3) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

(4) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

**18.** Which is the correct order of ionic sizes $_{63}Eu$, $_{71}Lu$, $_{70}Yb$ & $_{64}Gd$

(1) Gd > Eu > Yb > Lu

(2) Eu > Gd > Yb > Lu

(3) Eu > Gd > Lu > Yb

(4) Lu > Yb > Gd > Eu

**19.** Which of the following process requires maximum amount of energy ?

(1) $M^-_{(g)} \rightarrow M_{(g)}$ (2) $M_{(g)} \rightarrow M^+_{(g)}$

(3) $M^+_{(g)} \rightarrow M^{+2}_{(g)}$ (4) $M^{2+}_{(g)} \rightarrow M^{+3}_{(g)}$

**20.** If the configuration of $Y^+$ & $X^-$ is $2s^2\ 2p^6$ then

(1) IP of X = IP of Y

(2) EA of X > EA of Y

(3) IP of $X^-$ = IP of $Y^+$

(4) IP of $X^-$ > IP of $Y^+$

**21.** Match the column :

| Element (outer most shell $e^-$ configuration) | | Successive value of I.E. (eV/atom) | |
|---|---|---|---|
| (A) | $3s^1$ | (P) | 22, 25, 36, 42, 250 |
| (B) | $3s^2$ | (Q) | 18, 26, 32, 240, 260 |
| (C) | $3s^23p^1$ | (R) | 20, 24, 210, 230 |
| (D) | $3s^23p^2$ | (S) | 15, 180, 200, 220 |

Match the type of element, with the corresponding successive value of I.E :

(1) A-S, B-P, C-Q, D-R

(2) A-R, B-S, C-P, D-Q

(3) A-S, B-R, C-Q, D-P

(4) A-S, B-R, C-P, D-Q

**22.** Which of the following E.A. order is **not correct**?

(1) $N < O < S$

(2) $Cl > O > N > C$

(3) $O < S < F < Cl$

(4) $I < Br < F < Cl$

**23.** Which of the following process is non spontaneous?

(1) $F^+_{(g)} + Cl_{(g)} \rightarrow F_{(g)} + Cl^+_{(g)}$

(2) $O^-_{(g)} + S_{(g)} \rightarrow O_{(g)} + S^-_{(g)}$

(3) $Be^-_{(g)} + B_{(g)} \rightarrow Be_{(g)} + B^-_{(g)}$

(4) $N_{(g)} + O^+_{(g)} \rightarrow N^+_{(g)} + O_{(g)}$

**24.** **Statement-1 :** Electron gain enthalpy of N is +ve while that of P is −ve.

**Statement-2 :** Smaller atomic size of N in which there is a considerable electron-electron repulsion and hence the additional electron is not accepted easily.

(1) Both statements are true and the statement-2 is the correct explanation of statement-1.

(2) Both statements are true but statement-2 is not the correct explanation of statement-1.

(3) Statement-1 is true but the statement-2 is false.

(4) Statement-1 is false but the statement-2 is true

**25.** Identify the incorrect statement :-

(1) He has the highest $IE_1$ in periodic table

(2) Hg & $Br_2$ are liquid at room temperature

(3) F has highest electron affinity

(4) Zn, Cd, Hg are most volatile metals of periodic table.

**26.** Select correct order of given property :-

(1) $HOF < HOCl < HOBr < HOI$ Acidic nature

(2) $HF > HCl > HBr > HI$ Acidic nature

(3) $N_2O < N_2O_3 < N_2O_4 < N_2O_5$ Acidic nature

(4) $H_2O > H_2S > H_2Se > H_2Te$ Acidic nature

**27.** Identify incorrect order :-

(1) $Mg^{+2} < K^+ < S^{-2} < Se^{-2}$ ; Ionic radius

(2) $F^-(aq) > Cl^-(aq) > Br^-(aq) > I^-(aq)$ ; Electrical conductance

(3) $Cl > F > Br > I$ ; Electron Affinity

(4) $Li^+ < Mg^{+2} < Al^{+3}$ ; Hydration Energy

**28.** Which is wrong statement?

(1) The decreasing order of bond angle is $H_2O > H_2S > H_2Se > H_2Te$

(2) The decreasing order of bond angle is $NH_3 > PH_3 > AsH_3 > SbH_3$

(3) The decreasing order of bond dissociation energy is $Cl_2 > F_2 > Br_2 > I_2$

(4) The decreasing order of bond angle is $CH_4 > NH_3 > H_2O$

**29.** Which of the following combination of orbitals does not form covalent bond (x-axis is internuclear axis):-

(1) $s + p_y$

(2) $p_y + p_y$

(3) $d_{yz} + d_{yz}$

(4) $d_{xy} + d_{xy}$

**30.** **Assertion :** At room temperature $O_2$ and $N_2$ are stable molecule but $S_2$ and $P_2$ are less stable.

**Reason :** $2p\pi - 2p\pi$ collateral overlapping is effective but $3p\pi - 3p\pi$ overlapping is weak.

(1) Both assertion and reason are correct and reason is the correct explanation of assertion.

(2) Both assertion and reason are correct but reason is not correct explanation of assertion.

(3) Assertion is correct, reason is incorrect

(4) Both assertion and reason are incorrect.

**31.** Consider the following table :-

| Steric no. ($\sigma$ bp + $\ell$.p.) | $\ell$.p. on central atom | Shape of molecule |
|---|---|---|
| 5 | a | Linear |
| b | 1 | See-Saw |
| 6 | c | Square Planar |
| d | 2 | T-Shaped |

The value of a + b + c + d is :-

(1) 10 (2) 9 (3) 15 (4) 8

**32.** Which of the following do not have coordinate bond ?

(1) $PH_3$

(2) $P_2H_6^{+2}$

(3) $P_2H_5^{\oplus}$

(4) $PH_4^+$

**33.** Which of the order is correct ?

(1) $ClO^{\ominus} < ClO_2^{\ominus} < ClO_3^{\ominus}$ (Bond order)

(2) $CO_3^{-2} > CO > CO_2$ (Bond strength)

(3) $PO_4^{-3} > SO_4^{-2}$ (no. of coordinate bonds)

(4) $C_3O_2 > C(CN)_4$ (no. of $\pi$ bond)

**34.** **Statement-1 ($S_1$) :** Dipole moment of $CH_3Cl$ is more than $CH_3F$.

**Statement-2 ($S_2$) :** Dipole moment depends upon difference in electronegativity of C – X bond only

(1) Both $S_1$ and $S_2$ are correct

(2) $S_1$ is correct but $S_2$ is incorrect

(3) Both $S_1$ and $S_2$ are incorrect

(4) $S_1$ is incorrect but $S_2$ is correct.

**35.** Which of the following is the correct order of dipole moment ?

(1) $NH_3 < BF_3 < NF_3 < H_2O$

(2) $BF_3 < NF_3 < NH_3 < H_2O$

(3) $BF_3 < NH_3 < NF_3 < H_2O$

(4) $H_2O < NF_3 < NH_3 < BF_3$

**36.** Correct order of boiling point is :

(1) $H_2O < CCl_4 < CS_2 < CO_2$

(2) $CO_2 < CS_2 < CCl_4 < H_2O$

(3) $CS_2 < H_2O < CO_2 < CCl_4$

(4) $CCl_4 < H_2O < CO_2 < CS_2$

**37.** Which of the following statement is incorrect ?

(1) The magnitude of the H-bonding depends on the physical state of the compound

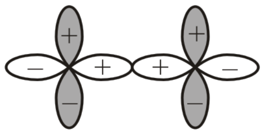
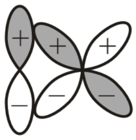
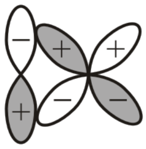
(2) $.....\underset{x}{H-F}\underset{y}{.....}H-F.....H-F.....$

x & y are bond lengths then x > y

(3) H-bond strength - $HF > H_2O > NH_3$

(4) Hydrogen bonds have strong influence on the structure and properties of the compounds.

**38.** Match the orbital overlap figures shown in List-I with the description given in List-II and select the correct answer using the code given below the lists.

| | List-I | | List-II |
|---|---|---|---|
| (P) |  | (1) | p – d ($\pi$ antibonding) |
| (Q) |  | (2) | d – d ($\sigma$ bonding) |
| (R) |  | (3) | p – d ($\pi$ bonding) |
| (S) |  | (4) | s – s ($\sigma$ antibonding) |

**Code :**

| | P | Q | R | S |
|---|---|---|---|---|
| (1) | 2 | 1 | 3 | 4 |
| (2) | 4 | 3 | 1 | 2 |
| (3) | 2 | 3 | 1 | 4 |
| (4) | 4 | 1 | 3 | 2 |

**39.** Which of the following order is incorrect :-

(1) $O_2^+ > O_2^-$ Number of electrons in BMO

(2) $B_2 > C_2$ Paramagnetism

(3) $B_2 < C_2$ Bond order

(4) $NO_2^+ > NO_2^-$ Bond angle

**40.** Order of covalent character is not correct for :-

(1) $NaF < Na_2O < Na_3N$

(2) $PbO < PbS$

(3) $CdCl_2 < CaCl_2$

(4) $AgCl > KCl$

**41.** Dipole-dipole (Keesom) attraction is present in :-

(1) $KCl + H_2O$

(2) $CH_3-\underset{\underset{O}{\|}}{C}-CH_3 + CH_3-C\equiv N$

(3) $NO_3^- + Cl_2$

(4) $HCl + Cl_2$

**42.** Pick number of species which can exist ?
$PH_5$, $XeF_5^{\ominus}$, $SiF_6^{-2}$, $PI_6^-$, $SBr_6$, $F_3^-$, $ClF_7$

(1) 1 (2) 2 (3) 3 (4) 4

**43.** The radius and charge of each of six ions are shown in the tables.

| ion | $K^+$ | $L^+$ | $M^+$ | $X^-$ | $Y^-$ | $Z^{-2}$ |
|---|---|---|---|---|---|---|
| radius/nm | 0.14 | 0.18 | 0.15 | 0.14 | 0.18 | 0.15 |

The ionic solids KX, LY and $M_2Z$ are of the same lattice type. What is the correct order of their lattice energies placing the one with the highest numerical value first :-

(1) $KX > LY > M_2Z$ (2) $KX > M_2Z > LY$

(3) $LY > M_2Z > KX$ (4) $M_2Z > KX > LY$

**44.** Which of the following relation is not true :-

(1) $\vec{\mu} = q \times d$ (2) $F.C. = VE + \dfrac{\ell p(e^-)}{2}$

(3) $B.O. = \dfrac{N_b - N_a}{2}$ (4) All

**45.** Match the following :-

| | Bonds | | BE (kJ $mole^{-1}$) |
|---|---|---|---|
| (i) | O–O | (p) | 464 |
| (ii) | O–H | (q) | 138 |
| (iii) | S–S | (r) | 339 |
| (iv) | S–H | (s) | 213 |

The correct match is -

(1) i-q, ii-p, iii-s, iv-r (2) i-q, ii-p, iii-r, iv-s

(3) i-p, ii-q, iii-s, iv-r (4) i-r, ii-q, iii-p, iv-s

## SUBJECT : BIOLOGY

**Topic :** SYLLABUS-2.

**46.** DNA is not present in –

(1) Chloroplast (2) Ribosomes

(3) Nucleus (4) Mitochondria

**47.** Match the column-I with column-II and select correct option :-

| Column-I | | Column-II | |
|---|---|---|---|
| a | Cristae | I | Primary constriction in chromosome |
| b | Thylakoids | II | Disc shaped sacs in Golgi apparatus |
| c | Centromere | III | Infoldings in mitochondria |
| d | Cisternae | IV | Flattened membranous sacs in stroma of plastids |

(1) a – III, b – IV, c – I, d – II

(2) a – II, b – III, c – IV, d – I

(3) a – IV, b – III, c – II, d – I

(4) a – I, b – IV, c – III, d – II

**48.** Outermost covering is cell membrane in case of-

(1) Plant (2) Fungi

(3) Algae (4) Animal

**49.** Read the following statements and choose the correct option.

**Statement - I :** The centriole forms the axoneme of cilia and flagella and spindle fibres that give rise to spindle apparatus during cell division in plant cells.

**Statement - II :** The cytoskeleton in a cell is involved in many functions such as mechanical support, motility, maintenance of the shape of the cell.

(1) Both Statements are correct.

(2) Both Statements are incorrect.

(3) Statement-I is correct but statement-II is incorrect.

(4) Statement-I is incorrect but statement-II is correct.

**50.** Select the incorrect statements :

(a) Golgi apparatus is the important site for synthesis of lipid.

(b) The cell envelope in a bacterial cell consist of cell wall and plasma membrane only.

(c) Both the centrioles in a centrosome lie perpendicular to each other in which each has an organisation like the cartwheel.

(d) The hydrolytic enzymes of lysosomes are optimally active at basic pH.

(1) a, b and c (2) b, c and d

(3) a, b and d (4) a, c and d

**51.** Centriole consist of 9 evenly spaced peripheral ____fibrils of tubulin :

(1) Doublet

(2) Singlet

(3) Triplet

(4) Singlet and doublet

**52.** Which of the following statement is incorrect ?

(1) Nucleus as a cell organelle was first described by Robert brown.

(2) Material of nucleus is stained by acidic dye.

(3) Chromatin term was given by flemming.

(4) Normally, there is only one nucleus per cell.

**53.** Algal cell wall is made up of-

(1) Cellulose, Galactans, Hemicellulose, Pectins

(2) Cellulose, Hemicellulose, Pectins, Protein

(3) Cellulose, Galactans, Mannans & Minerals likes $CaCO_3$

(4) Cellulose, Hemicellulose, Galactans, Mannans

54. Which among the following holds or glues the different neighbouring plant cells together ?

(1) Primary wall

(2) Secondary wall

(3) Tertiary wall

(4) Middle lamella

55. Which type of plastid is involved in storage of starch?

(1) Chloroplast

(2) Elaioplast

(3) Chromoplast

(4) Amyloplast

56. No organelles like the eukaryotes are found in prokaryotes except -

(1) Nucleus

(2) Lysosome

(3) Golgibody

(4) Ribosome

57. Chlorophyll pigments are present in which part of the chloroplast ?

(1) Stroma

(2) Thylakoid

(3) Outer chloroplastidial membrane

(4) Both (1) and (2)

58. Microbodies are :

(1) Membrane bound minute vesicles

(2) Contain various enzymes

(3) Present in both plant and animal cells

(4) All of these

59. Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R).

**Assertion (A) :** Anaphase-II begins with the simultaneous splitting of the centromere of each chromosome allowing them to move toward opposite poles of the cell.

**Reason (R) :** The shortening of microtubules attached to kinetochore takes place during anaphase-II.

Choose the correct answer from the options given below :

(1) Both (A) and (R) are true but (R) is not the correct explaination of (A).

(2) (A) is true but (R) is false.

(3) (A) is false but (R) is true.

(4) Both (A) and (R) are true and (R) is correct explaination of (A).

60. Which of the following is true for telophase-I ?

(1) Nuclear membrane reappears

(2) Nucleolus disappears

(3) Chromatin condensed into chromosome

(4) Each chromosome has one chromatid

61. Match the following-

| | Column-I | | Column-II |
|---|---|---|---|
| (a) | Prophase-I | (i) | Bivalent chromosomes align on equatorial plate |
| (b) | Metaphase-I | (ii) | Homologous chromosome get separated |
| (c) | Anaphase-I | (iii) | Synaptonemal complex formation |
| (d) | Telophase-I | (iv) | Nucleolus reappear |

(1) (a–i), (b–ii), (c–iii), (d-iv)

(2) (a–iii), (b–ii), (c–iv), (d-i)

(3) (a–iii), (b–i), (c–ii), (d-iv)

(4) (a–iv), (b–iii), (c–ii), (d-i)

**62.** A very significant contribution of mitosis is-

(1) Cell repair

(2) Reduction in number of chromosomes

(3) Increase genetic variability

(4) Recombination

**63.** A diploid cell contain 8 chromosomes then what will be number of chromosome in each daughter nuclei after meiosis-I ?

(1) 8 (2) 4 (3) 6 (4) 16

**64.** **Assertion :** Quiescent stage ($G_0$) of cell cycle is called inactive stage.

**Reason :** Cells in $G_0$ stage remain metabolically active but no longer proliferate.

(1) Both Assertion and Reason are correct and Reason is correct explanation of Assertion.

(2) Assertion is correct but Reason is incorrect.

(3) Assertion is incorrect but Reason is correct.

(4) Both Assertion and Reason are correct but Reason is not the correct explanation of Assertion.

**65.** Which process involves division of cells resulting in formation of two identical daughter cells ?

(1) Meiosis (2) Mitosis

(3) Both 1 and 2 (4) Meiosis-I

**66.** Which of the following statement is correct for mitotic metaphase ?

(1) Nucleolus, Golgi complex and ER reform

(2) Chromosome cluster at opposite spindle poles and their identity is lost as discrete elements

(3) Chromosomal material start to condense to initiate formation of distinct chromosome.

(4) Chromosome are moved to spindle equator and get aligned along metaphase plate through spindle fibers to both poles.

**67.** The complete disintegration of __A__ marks the start of __B__ of mitosis, hence the chromosomes are spread through the cytoplasm of the cell. Identify A and B.

(1) A–Cell membrane, B–Prophase

(2) A–Nuclear membrane, B–Prophase

(3) A–Nuclear membrane, B–Metaphase

(4) A–Cell membrane, B–Metaphase

**68.** Which stage of mitosis follows S and $G_2$ phase of interphase?

(1) Prophase

(2) Metaphase

(3) Anaphase

(4) Telophase

**69.** Find out the incorrect statement-

(1) In zygotene, formation of synaptonemal complex takes place

(2) In pachytene, formation of recombination nodules takes place

(3) In diakinesis, dissolution of synaptonemal complex takes place

(4) Crossing over takes place during pachytene

**70.** In which phase of cell cycle, synthesis of histone protein takes place ?

(1) $G_1$ phase

(2) S phase

(3) $G_2$ phase

(4) $G_0$ phase

**71.** Fill up the blanks in the following statements with the appropriate options.

(I) The complex formed by a pair of synapsed homologous chromosomes is called as <u>A</u> .

(II) Enzyme involved in crossing over is <u>B</u> .

(III) The X-shaped structure formed during diplotene is <u>C</u> .

(IV) Dyad of cells are formed at the end of <u>D</u> .

| | A | B | C | D |
|---|---|---|---|---|
| (1) | Synapsis | Permease | Synaptonemal complex | Telophase-I |
| (2) | Bivalent | Recombinase | Chiasmata | Meiosis-I |
| (3) | Tetrad | Lyase | Bivalent | Meiosis-I |
| (4) | Nodules | Endonucleas | Chiasmata | Anaphase-I |

**72.** In a diploid cell after S-phase quantity of DNA is 20 pico gram then after meiosis-II each daughter nucleus will be :-

(1) Diploid with 10 pico gram DNA

(2) Haploid with 10 pico gram DNA

(3) Haploid with 5 pico gram DNA

(4) Diploid with 5 pico gram DNA

**73.** Yeast can progress through the cell cycle in only about -

(1) 60 minutes

(2) 30 minutes

(3) 120 minutes

(4) 90 minutes

**74.** Given Below are two statements :-

**Statement - I :** A protein is imagined as a line, the left end represented by first amino acid (N - terminal) and right end represented by last amino acid (C - terminal).

**Statement - II :** Adult human haemoglobein, consists of four subunits (two sub units of $\alpha$-type and two subunits of $\beta$-types)

In the light of above statements, choose the correct answer from the options given below.

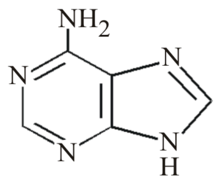
(1) Statement-I and II both are correct.

(2) Statement-I and II both are incorrect.

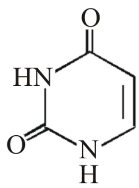
(3) Only Statement-I is correct.

(4) Only Statement-II is correct.

**75.**




(A)




(B)

(A) and (B) in the above diagram are :-

| | (A) | (B) |
|---|---|---|
| (1) | Guanine | Cytosine |
| (2) | Adenine | Uracil |
| (3) | Adenosine | Uridine |
| (4) | Guanine | Uracil |

**76.** Choose the odd one out with respect to homopolymers :

(1) Cellulose (2) Glycogen

(3) Protein (4) Chitin

**77.** Which of the following is considered as the simplest amino acid -

(1) Alanine (2) Glycine

(3) Arginine (4) Tryptophane

**78.** Which of the following is not a protein?

(1) GLUT-4 (2) Collagen

(3) Inulin (4) Antibody

**79.** Which of the following pair is incorrect regarding secondary metabolites ?
(1) Pigments – Anthocyanins
(2) Essential oil – Lemon grass oil
(3) Toxins – Curcumin
(4) Drugs – Vinblastin

**80.** Arachidonic acid and palmitic acid has how many carbon atoms in their structure respectively ?
(1) 19, 15 (2) 15, 19 (3) 20, 16 (4) 16, 20

**81.** Which functional group is found in glucose ?
(1) Aldehyde (2) Ketone
(3) Alcohol (4) Ether

**82.** Which of the following is linear polymer of fructose ?
(1) Chitin (2) Cellulose
(3) Pectins (4) Inulin

**83.** Codeine is a :-
(1) Toxins (2) Drug
(3) Alkaloid (4) Polymeric substance

**84.** Substrate binds at which site of enzyme ?
(1) Allosteric site (2) Inhibitory site
(3) Active site (4) None of these

**85.** The class which includes all the enzymes catalysing inter-conversion of optical, geometric or positional isomers is called as :
(1) Ligases (2) Isomerases
(3) Hydrolases (4) Transferases

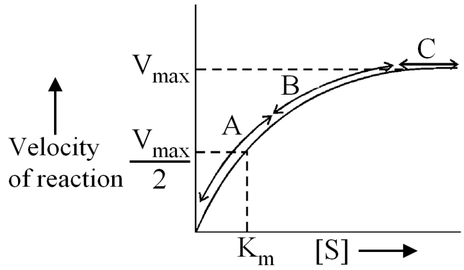
**86.** Select the correct statements :
(a) Some nucleic acid that behave like enzyme are called ribozyme.
(b) Proteins with catalytic power are named enzymes.
(c) Prosthetic groups are only organic compounds which are loosely bound to apoenzyme.
(d) Catalytic activity of an enzyme is lost when co-factor is removed
(1) a and d (2) b, c and d
(3) a, b and d (4) a and d

**87.** ________ is a metal ion cofactor for the proteolytic enzyme carboxypeptidase.
(1) Mn (2) Zn
(3) Cl (4) Mg

**88.** Which of the following co-factor is a prosthetic group ?
(1) $NAD^+$ (2) Zinc
(3) Haem (4) Niacin

**89.** Essential chemical component of NAD and NADP is :-
(1) Zinc (2) Haem
(3) Niacin (4) $Mg^{2+}$

**90.** Which of the following statement is not correct regarding relationship between substrate concentration and rate of enzyme catalysed reaction as depicted in the graph :-




(1) $K_m$ is measure of catalytic activity of an enzyme.
(2) With increase in substrate concentration, the velocity of enzymatic reactions rise in segment 'A' of the curve
(3) Enzymatic reaction velocity remains constant in segment 'B' of the curve with the increase in substrate concentration.
(4) In segment 'C' of the curve, the enzyme molecules are fewer than the substrate molecules and after saturation of these molecules, there are no free enzyme molecules to bind with the additional substrate molecules.

**91.** Fibrins are formed by the conversion of inactive fibrinogens in the plasma by the enzyme _____.

(1) Thrombin (2) Thromboplastin

(3) Thrombokinase (4) Prothrombin

**92.** Read the following statements and select the correct option.

**Statement-A :** In each cardiac cycle two prominent sounds are produced which can be easily heard through a stethoscope.

**Statement-B :** First heart sound is (lub) associated with closure of semilunar valves.

(1) Both Statement are correct & Statement-B is correct explanation of Statement-A.

(2) Both Statement are correct but Statement-B is not a correct explanation of Statement-A.

(3) Only Statement-A is correct.

(4) Only Statement-B is correct.

**93.** Which of the following can increase cardiac output?

(1) Sympathetic nerve

(2) Adrenal medullary hormones

(3) Both (1) & (2)

(4) Thyrocalcitonin

**94.** Read the following statements & select the correct option.

**Statement-A :** Lymph is a coloured fluid containing specialised lymphocytes which are responsible for immune response of the body.

**Statement-B :** Lymph is also important carrier for nutrients, hormones etc.

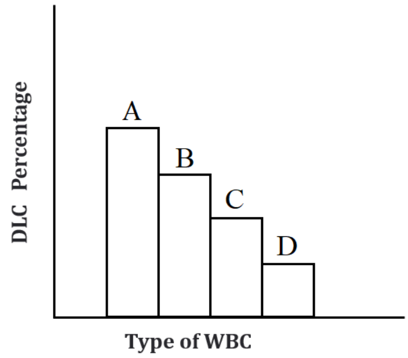
(1) Both Statement are correct & Statement-B is correct explanation of Statement-A.

(2) Both Statement are correct but Statement-B is not a correct explanation of Statement-A.

(3) Only Statement-A is correct.

(4) Only Statement-B is correct.

**95.**




Recognise A,B, C,D

(1) A-Neutrophils, B-Lymphocytes, C-Basophils, D-Monocyte

(2) A-Neutrophils, B-Lymphocytes, C-Monocyte, D-Acidophils

(3) A-Lymphocytes, B-Neutrophils, C-Acidophils, D-Basophils

(4) A-Monocytes, B-Neutrophils, C-Lymphocytes, D-Eosinophils

**96.** Which of the following cells are phagocytic in nature ?

(1) Acidophils

(2) Neutrophils

(3) Monocytes

(4) Both (2) & (3)

**97.** Read the following statements carefully and choose the correct option :-

**Statement-1 :-** A reduction in number of thromobocytes can lead to clotting disorders.

**Statement-2 :-** Megakaryocytes are fragments of thrombocytes which are involved in coagulation.

(1) Both the statement-1 and 2 are correct

(2) Statement-1 is correct but statement-2 is incorrect

(3) Statement-2 is correct but statement-1 is incorrect

(4) Both the statements-1 and 2 are incorrect

**98.** Incorrect statement about A,B and Rh antigens ?

(1) Present on the surface of RBCs

(2) Can induce immune response

(3) Anti-B antibodies destroys antigen A

(4) Can stimulate the production of antibodies

**99.** A ___A___ person if exposed to ___B___ blood, will form specific antibodies against Rh antigen :-

(1) A ⟶ Rh –ve, B ⟶ Rh +ve

(2) A ⟶ Rh +ve, B ⟶ Rh –ve

(3) A ⟶ Rh +ve, B ⟶ Rh +ve

(4) A ⟶ Rh –ve, B ⟶ Rh –ve

**100.** Which of the following is/are absent in lymph ?

(1) RBCs

(2) WBCs

(3) Larger proteins

(4) Both (1) & (3)

**101.** Opening of bicuspid and tricuspid valves occurs at :-

(1) Starting of atrial contraction

(2) Starting of ventricular contraction

(3) Starting of joint diastole

(4) Starting of atrial diastole

**102.** **Assertion (A) :** $CO_2$ is converted to bicarbonate at tissue level & transported to the alveoli is released out as $CO_2$.

**Reason (R) :** Carbonic anhydrase facilitates the trapping of $CO_2$

(1) Both Assertion and Reason are false

(2) Assertion and Reason both are true but reason is not correct explanation of assertion

(3) Both assertion & reason are true and reason is correct explanation of assertion

(4) Assertion is true but Reason is false

**103.** Globulins contained in human blood plasma are primarily involved in :-

(1) Clotting of blood (2) Osmotic pressure

(3) Defence mechanism (4) Oxygen transport

**104.** Which of the following statements are **correct** ?

(A) A fall in glomerular blood flow can activate the JG cells to release renin.

(B) Angiotensin II is powerful vasodialator.

(C) Angiotensin II also activates the adrenal medulla to release aldosterone.

(D) Aldosterone causes reabsorption of $Na^+$ and water from the distal parts of the tubule.

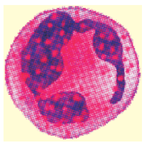
(E) Renin converts angiotensinogen in blood to angiotensin I and further to angiotensin II.

Choose the correct answer from the options given below :

(1) B, C and D only (2) A, B and E only

(3) A, D and E only (4) A, C and D only

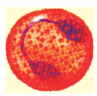
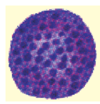
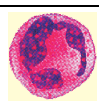
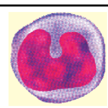
**105.** Identify the cell shown in the diagram :-



(1) Neutrophil (2) Basophil

(3) Eosinophil (4) Monocytes

**106.** Find out the correct match from the table given below :

| | Cells | Name | Percent | Function |
|---|---|---|---|---|
| 1. |  | Eosinophils | 6–8% | Associated with allergic reactions |
| 2. |  | Basophils | 0.5–1% | Inflammatory reaction |
| 3. |  | Neutrophils | 60–65% | Heparin Secretion |
| 4. |  | Monocytes | 2–3% | Phagocytosis |

**107.** Identify the blood groups shown in the diagram.

| Red blood cell type | (i) | (ii) | (iii) | (iv) |
|---|---|---|---|---|
| Antibodies in plasma | Anti-B | Anti-A | None | Anti-A and Anti-B |

| | (i) | (ii) | (iii) | (iv) |
|---|---|---|---|---|
| 1. | A | O | AB | B |
| 2. | A | B | AB | O |
| 3. | A | B | O | AB |
| 4. | A | AB | B | O |

**108.** In which of the following animals, respiration occurs through lungs ?

(1) Bird (2) Fish

(3) Cockroach (4) Earthworm

**109.** During swallowing glottis can be covered by _______ to prevent the entry of food into larynx.

(1) Pharynx (2) Lips

(3) Epiglottis (4) Alveoli

**110.** In human respiration takes place in

(1) Cells lining the lungs cavity

(2) Cells found in blood

(3) All living cells of the body

(4) Only RBC

**111.** Primary site for exchange of gases in human is-

(1) Nostrils (2) Nasal passage

(3) Pharynx (4) Alveoli

**112.** $PO_2$ in deoxygenated blood is-

(1) 40 mmHg (2) 95 mmHg

(3) 45 mmHg (4) 159 mmHg

**113.** When you hold your breath, which of the following gas changes in blood would first lead to the urge to breathe?

(1) rising $CO_2$ concentration

(2) falling $CO_2$ concentration

(3) rising $CO_2$ and falling $O_2$ concentration

(4) falling $O_2$ concentration

**114.** Select the correct statement.

(1) Expiration occurs due to contraction of external intercostal muscles

(2) Intrapulmonary pressure is lower than the atmospheric pressure during inspiration.

(3) Inspiration occurs when atmospheric pressure is less than intrapulmonary pressure.

(4) Expiration is initiated due to contraction of diaphragm.

**115.** TV + ERV is

(1) RV (2) IC

(3) EC (4) FRC

**116.** Read the following statements :-

(a) Respiratory rhythm centre present in pons region of brain

(b) The role of oxygen in the regulation of respiratory rhythm is quite insignificant

(c) Receptors associated with aortic arch and carotid artery can recognise changes in $CO_2$ and $H^{\oplus}$ Concentration.

(d) Pneumotaxic centre can reduce the duration of inspiration and thereby alter the respiratory rate.

How many statements are correct about regulation of Respiration ?

(1) One

(2) Four

(3) Two

(4) Three

**117.** Find out **correct** statement :-

(1) Inspiration is a passive process

(2) Inspiration is an active process

(3) Expiration is an active process

(4) Both expiration and inspiration are passive process

**118.** Match the following columns.

| Column-I | | Column-II | |
|---|---|---|---|
| A | Earthworms | I | Tracheal-Tubes |
| B | Aquatic arthropods | II | Lungs |
| C | Molluscs | III | Gills |
| D | Reptiles | IV | Moist cuticle |
| E | Insects | | |

(1) A= I, B= II, C= III, D= IV, E= III

(2) A= IV, B= III, C= III, D= II, E= I

(3) A= IV, B= II, C= I, D= III, E= III

(4) A= IV, B= II, C= III, D= I, E= III

**119.** Match the items in column I with column II and choose the correct option :-

| | Column-I | | Column-II |
|---|---|---|---|
| (A) | 2500 to 3000 ml | (i) | Tidal volume |
| (B) | 1000 - 1100 ml | (ii) | Inspiratory reserve volume |
| (C) | 500 ml | (iii) | expiratory reserve volume |
| (D) | 5100 - 5800 ml | (iv) | Total lung capacity |
| (E) | 1100 - 1200 ml | (v) | Residual volume |

| | A | B | C | D | E |
|---|---|---|---|---|---|
| (1) | (ii) | (iv) | (i) | (iii) | (v) |
| (2) | (ii) | (iii) | (i) | (v) | (iv) |
| (3) | (ii) | (iii) | (i) | (iv) | (v) |
| (4) | (ii) | (v) | (i) | (iv) | (iii) |

**120.** "Chemoreceptors" which recognise the change concentration of $CO_2$ and $H^+$ are found in -

(1) Aortic arch (2) Carotid artery

(3) Medulla (4) All of these

**121.** The vital capacity of human is about :-

(1) 2500 ml (2) 500 ml

(3) 4100 ml (4) 6000 ml

**122.** Fibrous lungs can be seen in-

(1) Asthma

(2) Emphysema

(3) Occupational respiratory disorders

(4) Flu

**123.** Which of the following volume is not included in vital capacity ?

(1) ERV (2) TV

(3) IRV (4) RV

**124.** Which of the following statement is not correct about excretion?

(1) Ascending limb of Loop of Henle is impermeable to water

(2) On an average, human excretes 25 to 30 gm urea per day

(3) Glucose and amino acids are reabsorbed passively in renal tubule.

(4) 70-80 percent of electrolytes and water are reabsorbed in PCT

**125.** Which of the following statement is/are **true**?

(I) Glomerular filtrate is isotonic to plasma.

(II) When the urine passes into collecting duct, it becomes hypotonic.

(III) Filtrate is isotonic in proximal convoluted tubule.

(1) I and II (2) II and III

(3) I and III (4) All

**126.** The peritubular capillaries of the nephron arise from the :-

(1) Afferent arterioles (2) Renal artery

(3) Efferent arterioles (4) Vasa Recta

**127.** Which of the following part of nephron is found in medulla of kidney?

(1) Malpighian corpuscle

(2) PCT

(3) DCT

(4) Loop of Henle

**128.** Which of the following structure have stretch receptors for micturition?

(1) Urinary bladder (2) Ureters

(3) Renal pelvis (4) Column of Bertini

**129.** In the given how many are the secretions of sebaceous glands ?

Sterols, Lactic acid, Hydrocarbons, Waxes, Urea

(1) Five (2) Four (3) Three (4) Two

**130.** Match the abnormal conditions given in Column I with their explanations given in Column II and choose the correct option :-

| Column I | | Column II | |
|---|---|---|---|
| (A) | Glycosuria | (i) | Accumulation of uric acid in joints |
| (B) | Renal calculi | (ii) | Inflammation in Glomerulus |
| (C) | Glomerulo nephritis | (iii) | Mass of crystallised salts within the kidney |
| (D) | Gout | (iv) | Presence of glucose in urine |

(1) A–i, B–iii, C–ii, D–iv

(2) A–iii, B–ii, C–iv, D–i

(3) A–iv, B–iii, C–ii, D–i

(4) A–iv, B–ii, C–iii, D–i

**131.** **Assertion :-** Analysis of urine helps in clinical diagnosis of malfunctioning of kidney.

**Reason :-** Glucosuria is indicative of diabetes mellitus.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

**132.** In interstitial fluid of kidney an increasing osmolarity from cortex to the inner medulla. This gradient is mainly caused by :-

(1) $Na^+$, K

(2) NaCl, $H_2O$, $HCO_3$

(3) NaCl and Urea

(4) Urea, $HCO_3$

**133.** Which segment does not play a role in the maintenance of pH and ionic balance of blood by selective secretion :-

(1) PCT

(2) Bowman's capsule

(3) DCT

(4) Collecting duct

**134.** In human, ammonia produced by metabolism is converted in urea in the-

(1) Liver (2) Kidney

(3) Blood (4) Lung

**135.** Ammonia is the chief excretory substance in-

(1) Camel and whale

(2) Cartilaginous fishes

(3) Whale and tortoise

(4) Bony fishes

## SUBJECT : PHYSICS

**Topic :** SYLLABUS-2.

**136.** A two particle system of masses 2 kg and 3 kg present at (15, 0) and (0, 20) respectively in a xy plane then its centre of mass lies on the lines :-

(a) $y = 2x$ (b) $\frac{x}{12} + \frac{y}{24} = 1$

(c) $3x - y = 6$ (d) $y = x$

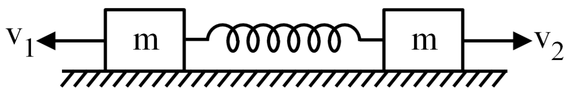
(1) Only (a) is correct

(2) Both (a) and (b) are correct

(3) (a), (b) and (c) are correct

(4) (b), (c) and (d) are correct

**137.** Two students were given a physics problem for finding maximum extension of spring if blocks are imparted velocities $v_1$ and $v_2$ when spring is unstretched.




**By student A :** $\frac{1}{2}m(v_1 + v_2)^2 = \frac{1}{2}kx^2$

**By student B :** $\frac{1}{2}mv_1^2 + \frac{1}{2}mv_2^2 = \frac{1}{2}kx^2$

(1) Student A is correct, student B is wrong.

(2) Student B is correct, student A is wrong.

(3) Both are correct.

(4) Both are wrong.

**138.** If velocities of 3 particles of mass 1 kg (each) in a system varies with time as $\vec{v}_1 = t^2\hat{i}$, $\vec{v}_2 = (t+1)\hat{j}$ and $\vec{v}_3 = \left(\frac{3}{2}t\hat{i} + t^2\hat{j}\right)$ respectively, then momentum of centre of mass of system at t = 2 sec is :-

(1) $7\sqrt{2}$ units (2) $4\sqrt{2}$ units

(3) $3\sqrt{2}$ units (4) $5\sqrt{2}$ units

**139.** Mass is distributed uniformly over a thin triangular plate and position of two vertices are given by (–2,3) and (–4,–2). What is the position of third vertex if centre of mass of the plate lies at the origin ?

(1) (6,1) (2) (1, –6)

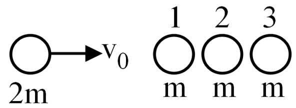
(3) (–6, 1) (4) (6, –1)

**140.** A bullet of mass 0.01 kg travelling at a speed of 500 m/s strikes a block of mass 2 kg which is suspended by a string of length 5m. The centre of gravity of block is found to rise a vertical distance of 0.2 m. Then find the speed of bullet with which it emerges out from the block ($g = 10\ m/s^2$) :-

(1) 200 m/s (2) 140 m/s

(3) 100 m/s (4) 280 m/s

**141.** A ball of mass 2m moving with a velocity $v_0$ collides head-on elastically with a cradle of three identical balls each of mass m. Determine velocity of ball 3 after collision :-




(1) $v_0/3$ (2) $v_0/2$ (3) $4v_0/3$ (4) $v_0$

**142.** A ball falls from a height 180 m upon a fixed horizontal plane and rebounds.($e = 0.5$, $g = 10\ m/s^2$).

(a) Velocity of ball after second collision is 25 m/s.

(b) Average speed for whole interval is $\frac{50}{3}$ m/s.

(c) Total distance travelled before rebounding has stopped is 300 m.

**Options :-**

(1) Only (a) is correct

(2) Only (a) and (b) are correct

(3) Only (b) and (c) are correct

(4) All (a), (b) and (c) are correct

**143.** A boy of mass 40 kg is standing in a gravity free space at a height 20m above the floor. He throws a ball of mass 0.2 kg downwards with a speed 5 m/s.

**Statement-1 :** When ball reaches the floor, the distance of the man above the floor will be 20.1 m.

**Statement-2 :** When ball reaches the floor the displacement of centre of mass of (boy + ball) system is 0.1 m.

(1) Only statement-1 is correct.

(2) Only statement-2 is correct.

(3) Both statement-1 and statement-2 are correct.

(4) Both statement-1 and statement-2 are wrong.

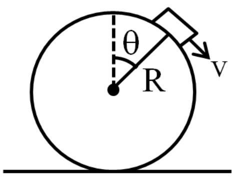
**144.** A ball is thrown vertically downwards with velocity $\sqrt{gh}$ from a height h above the ground after colliding with the ground, it just reaches at height h/2 above the ground. Coefficient of restitution is :-

(1) $\frac{1}{2}$ (2) 1 (3) $\frac{1}{\sqrt{2}}$ (4) $\frac{1}{\sqrt{3}}$

**145.** A particle of mass 1 kg describes a circle of radius 2 m. The centripetal acceleration of the particle is 8 $m/s^2$. What will be the momentum in $\left(\frac{kg-m}{sec}\right)$ of the particle ?

(1) 4 (2) 2 (3) 8 (4) 1

**146.** A small block of mass 'm' moves on the smooth surface of a fixed sphere as shown. Force applied by block on the sphere at the given instant is-




(1) $mg\cos\theta - \frac{mv^2}{R}$ radially outwards

(2) $mg\cos\theta - \frac{mv^2}{R}$ radially inwards

(3) $mg\sin\theta$

(4) None

**147.** A cyclist speeding 10 km/h on a level road takes a sharp circular turn of radius 3 m without reducing the speed. The coefficient of static friction between the road and the tyres is 0.1 then-

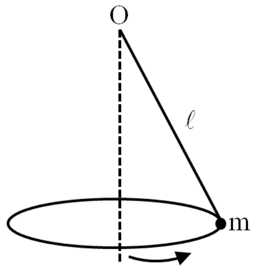
(1) He will slip

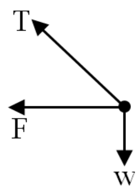
(2) He will not slip

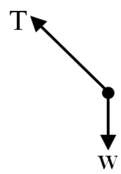
(3) both (1) and (2) can possible

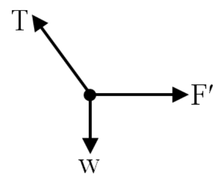
(4) Data insufficient

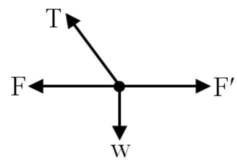
**148.** A point mass m is suspended from a light thread of length $\ell$, fixed at O, is whirled in a horizontal circle at constant speed as shown. From your point of view, stationary with respect to the mass, the forces on the mass are :-



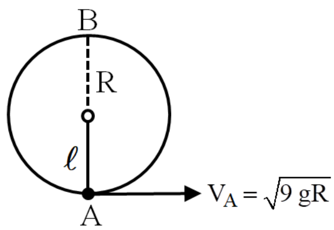

(1)



(2)



(3)



(4)



**149.** A stone is attached to one end of a string and is whirled in a vertical circle of radius R as shown in fig. Find the tension in string at highest point 'B' ? (m = mass of stone)




(1) $T_B = 6$ mg (2) $T_B = 4$ mg

(3) $T_B = 3$ mg (4) $T_B = 9$ mg

**150.** Match the following columns and choose the correct option from the codes given below for uniform circular motion.

| | Column-I | | Column-II |
|---|---|---|---|
| (A) | Speed | (1) | Constant |
| (B) | Velocity | (2) | Variable |
| (C) | Magnitude of acceleration | (3) | Zero |
| (D) | Acceleration | | |

(1) (A)-(1), (B)-(2), (C)-(2), (D)-(1)

(2) (A)-(1), (B)-(2), (C)-(1), (D)-(2)

(3) (A)-(2), (B)-(1), (C)-(2), (D)-(1)

(4) None

**151.** The speed of a particle moving in a circle of radius r = 2m varies with time t as $v = t^2$. The net acceleration at t = 2 sec is :-

(1) $\sqrt{40}$

(2) $\sqrt{80}$

(3) $\sqrt{90}$

(4) 8

**152.** **Statement-I :-** Torque is a vector whose direction is along the applied force

**Statement-II :-** $\vec{\tau} = \vec{r} \times \vec{F}$

(1) **Statement-I** is true, **Statement-II** is true and **Statement-II** is correct explanation of **Statement-I**

(2) **Statement-I** is true, **Statement-II** is true but **Statement-II** is not a correct explanation of **Statement-I**

(3) **Statement-I** is true, but **Statement-II** is false

(4) **Statement-I** is false, but **Statement-II** is true

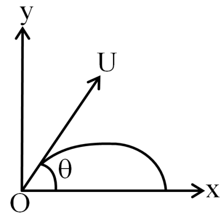
**153.**


Given uniform Rod is hinged about one end and other end is supported by string as shown in fig. If string is cut then angular acceleration of rod when it makes an angle of 37° from Horizontal

(1) $\alpha = \frac{6g}{L}$ (2) $\alpha = \frac{3g}{2L}$

(3) $\alpha = \frac{6g}{5L}$ (4) $\alpha = \frac{2g}{3L}$

**154.**




For the given Projectile of mass m the angular momentum about point of projection when particle is at maximum height is

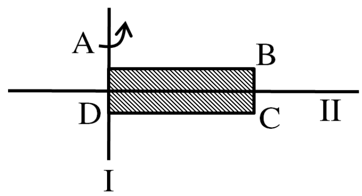
(1) $\frac{mU^3\sin^2\theta\cos\theta}{2g}$

(2) $\frac{mU^3\sin\theta\cos^2\theta}{g}$

(3) $\frac{mU^2\sin\theta^2\cos\theta}{g}$

(4) $mU\cos\theta$

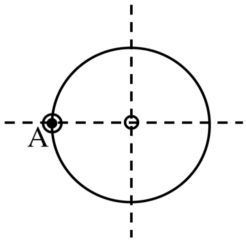
**155.** A rectangular slab ABCD have dimensions $\ell \times 2\ell$ as shown in figure. Then radius of gyration about axis-I is




(1) $\sqrt{\frac{2}{3}}\ell$ (2) $\frac{2\ell}{\sqrt{3}}$

(3) $\frac{\ell}{\sqrt{3}}$ (4) $\sqrt{3}\ell$

**156.** A thin wire of length $\ell$ and mass m is bent in the form of a circle as shown. Its moment of inertia about an axis passing through A and perpendicular to plane ?




(1) $\frac{m\ell^2}{\pi^2}$ (2) $m\ell^2$ (3) $\frac{2m\ell^2}{\pi^2}$ (4) $\frac{m\ell^2}{2\pi^2}$

**157.** Angular momentum of a particle rotating under a central force is constant due to :

(1) Constant torque

(2) Constant force

(3) Constant linear momentum

(4) Zero torque

**158.** A cylinder rolls on a horizontal plane surface. If the speed of the highest point is 50 m/s then speed of the centre is:

(1) 100 m/s (2) 25 m/s

(3) 12.5 m/s (4) $\frac{50}{\sqrt{2}}$ m/s

**159.** **Statement-1 :** For a point (other than points on the axis) on a rotating body, acceleration can't be zero.
**Statement-2 :** All points (other than points on the axis) on a rotating body moves on circular path.

(1) Both **Statement-1** and **Statement-2** are true and **Statement-2** is the correct explanation of **Statement-1**.

(2) Both **Statement-1** and **Statement-2** are true but **Statement-2** is NOT the correct explanation of **Statement-1**.

(3) **Statement-1** is true but **Statement-2** is false

(4) **Statement-1** is false but **Statement-2** is true.

**160.** Let $I_A$ and $I_B$ be moments of inertia of a body about two axes A & B respectively. The axis A passes through the centre of mass of the body but B does not then :

(1) $I_A < I_B$ always

(2) If the axes are parallel $I_A < I_B$

(3) If the axes are parallel $I_A = I_B$

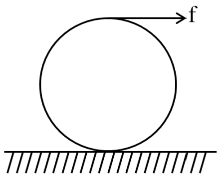
(4) If the axes are not parallel then $I_A \geq I_B$

**161.** The speed of center of mass of a homogenous sphere after rolling down an inclined plane through a vertical height h from rest without sliding is :

(1) $\sqrt{gh}$ (2) $\sqrt{gh/5}$

(3) $\sqrt{4gh/3}$ (4) $\sqrt{\frac{10gh}{7}}$

**162.** A disc of radius R and mass M is under pure rolling when a force f is applied at the top most point (as shown in fig.). If there is sufficient friction between the disc and the horizontal surface then :




(1) $a_{CM} = \frac{f}{M}$ (2) $a_{CM} < \frac{f}{M}$

(3) $a_{CM} > \frac{f}{M}$ (4) None

**163.** If density of earth is increase by 3% and radius is increase by 1% then percentage change in acceleration due to gravity at the surface of earth is :

(1) 4%

(2) 2%

(3) 3%

(4) No change

**164.** A point mass 'm' is placed inside a spherical shell of radius R and mass M at a distance of R/2 from centre of the shell. The gravitational force exerted by the shell on the point mass is :

(1) $\frac{GMm}{R^2}$ (2) $-\frac{GMm}{R^2}$

(3) Zero (4) $\frac{4GMm}{R^2}$

**165.** A body weighs 36 N on the surface of earth. When it is taken to a height R/2 from surface of earth, where R is radius of earth. Its weight would be ________

(1) 16 N (2) 28 N

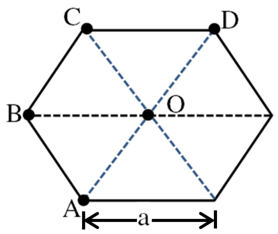
(3) 32 N (4) 72 N

**166.** Two particles of equal mass go round a circle of radius R under the action of their mutual gravitation attraction. Speed of each particle is :

(1) $\sqrt{\frac{Gm}{2R}}$ (2) $\sqrt{\frac{Gm}{4R}}$

(3) $\sqrt{\frac{Gm}{R}}$ (4) $\sqrt{\frac{2Gm}{R}}$

**167.** Four point masses each of mass m are placed at four corner A, B, C and D as shown. Gravitational field at O is :




(1) $\frac{\sqrt{3}Gm}{a^2}$ (2) $\frac{Gm}{\sqrt{3}a^2}$

(3) $\frac{\sqrt{3}Gm}{2a^2}$ (4) $\frac{2Gm}{a^2}$

**168.** The escape velocity from the earth is about 11 km/s. The escape velocity from a planet having twice the radius and the same mean density as the earth is :

(1) 22 km/s (2) 5.5 km/s

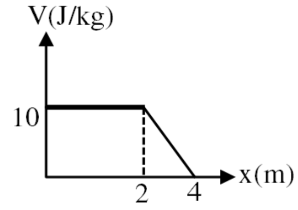
(3) 11 km/s (4) 15.5 km/s

**169.** Work done by an external agent in bringing three particles of mass m each at the vertices of an equilateral triangle of length '$\ell$' is :

(1) $\frac{3Gm^2}{\ell}$ (2) $\frac{3Gm^2}{2\ell}$

(3) $-\frac{3Gm^2}{\ell}$ (4) $-\frac{3Gm^2}{2\ell}$

**170.** Gravitational potential varies along x-axis as shown. Gravitational field at x = 3m is :




(1) $\vec{I} = -5\hat{i}$ (2) $\vec{I} = +5\hat{i}$

(3) zero (4) $\vec{I} = -2.5\hat{i}$

**171.** In a certain region of space, the potential is given by : $V = k[2x^2 - y^2 + z^2]$. The electric field at the point (1,1,1) has magnitude = :-

(1) $k\sqrt{6}$ (2) $2k\sqrt{6}$

(3) $2k\sqrt{3}$ (4) $4k\sqrt{3}$

**172.** A cube made of material having a density of $0.9 \times 10^3$ kg/m$^3$ floats between water and a liquid of density $0.8 \times 10^3$ kg/m$^3$, which is immiscible with water. What part of the cube is immersed in water ?

(1) $\frac{1}{2}$ (2) $\frac{2}{3}$

(3) $\frac{3}{4}$ (4) $\frac{3}{7}$

**173.** Sixty four spherical rain drops of equal size are falling vertically through air with a terminal velocity 1.5 ms$^{-1}$. If these drops coalesce to form a big spherical drop. then terminal velocity of big drop is:

(1) 8 ms$^{-1}$ (2) 16 ms$^{-1}$

(3) 24 ms$^{-1}$ (4) 32 ms$^{-1}$

**174.** A container of height 10 m is filled with water. An orifice is made on a side will of the container such that the water coming out from orifice can achieve maximum range. Then find the value of this maximum range.

(1) 5 m (2) $10\sqrt{2}$ m

(3) 10 m (4) 8 m

**175.** If a capillary tube is dipped into liquid and the levels of the liquid inside and outside are same then the angle of contact is -

(1) 0° (2) 30°

(3) 90° (4) 120°

**176.** Elastic potential energy density stored in a strained wire is U. If wire is replaced by identical wire of twice youngs modulus then elastic potential energy density for the same strain will be -

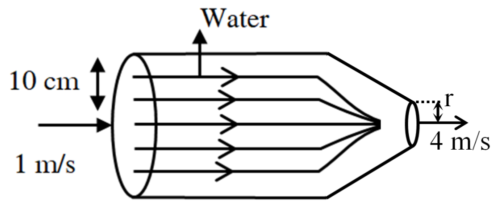
(1) 4U (2) 2U (3) 8U (4) 16U

**177.** The work done in increasing the size of a rectangular soap film with dimensions 8 cm × 3.75 cm to 10 cm × 6 cm is $2 \times 10^{-4}$ J. The surface tension of soap solution will be -

(1) $1.65 \times 10^{-2}$ N/m (2) $3.3 \times 10^{-2}$ N/m

(3) $6.6 \times 10^{-2}$ N/m (4) $8.25 \times 10^{-2}$ N/m

**178.** For adjoining fig, the value or r is -




(1) 5 cm (2) 10 cm

(3) 7 cm (4) None

**179.** A man is walking near a railway track then a train passes on the track. As train passes man feels a pull towards the train. Which concept describes this pull.

(1) Newton's Law of viscosity

(2) Stoke's Law

(3) Bernoulli's theorem

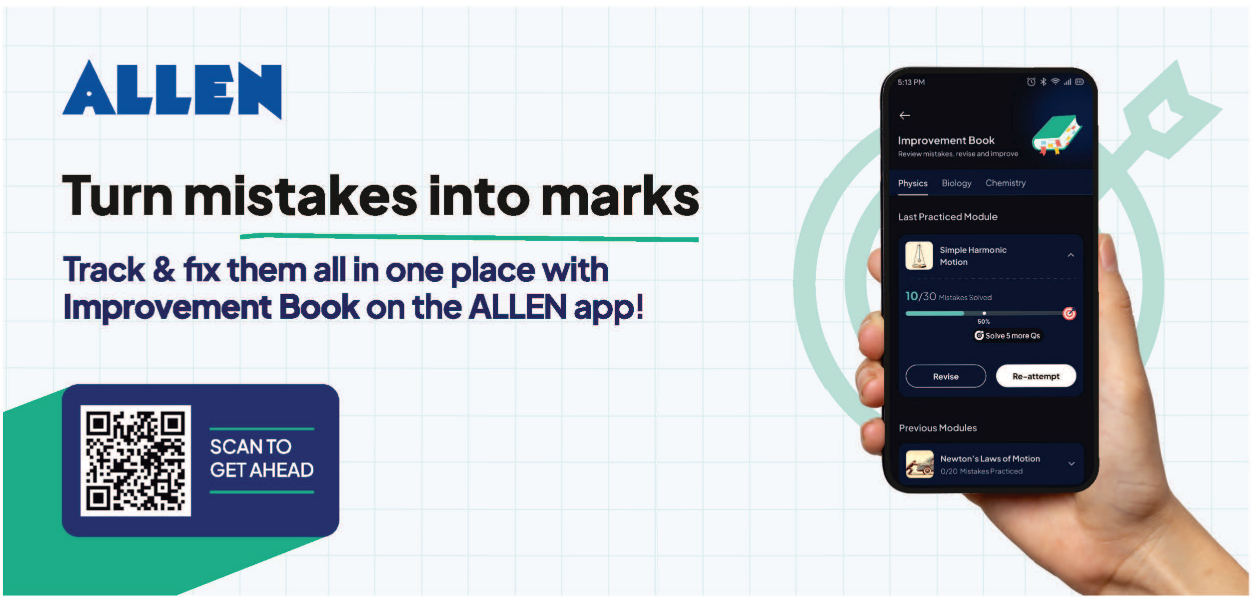
(4) Momentum conservation.

**180.** A fluid is moving through a horizontal pipe. The pressure at point 1 is $P_1$ = 4000 Pa and the velocity is 4 m/s. At point 2, the velocity is 2 m/s. What is the pressure at point 2? (Density of fluid = 800 kg/m$^3$)

(1) 4000 Pa (2) 4400 Pa

(3) 8000 Pa (4) 8800 Pa

SPACE FOR ROUGH WORK



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